FORMULA 1 TYRE DEGRADATION ESTIMATION USING ARIMA

A time-series forecasting study using lap-time data

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Project Report

Case study reported in this document: 2026 Singapore Grand Prix — Driver LEC
Selected stint: 3 | Tyre compound: SOFT | Clean laps: 22
Training observations: 18 | Held-out test observations: 4

Implementation: Python, FastF1, pandas, NumPy, Matplotlib, statsmodels
The source code is maintained separately in the associated GitHub repository.

Abstract

This work investigates whether an autoregressive integrated moving average (ARIMA) time-series model might forecast lap times in Formula 1 during a tyre stint with enough accuracy to guide estimations of tyre deterioration. Lap time is a quantifiable measure: although other elements including fuel load, other cars on the track, weather conditions, track changes, driver actions, pit choices, and general race conditions influence these times as well, a tyre's performance degrades and lap times could rise. Hence, the study aims to project performance changes related to a stint instead of determining the actual physical deterioration of the tyre. Using FastF1, the technique gathers race lap data; removes laps connected with pit lanes, non-green-flag events, or those that are erroneous; picks a certain stint; converts lap times into seconds; and saves the last portion of the chosen series for analysis. Using Akaike's Information Criterion (AIC), different ARIMA structures are investigated. Using root mean squared error (RMSE) and mean absolute error (MAE), the projections are contrasted with naive and linear-trend benchmarks. Augmented Dickey -Fuller (ADF) and Ljung–Box tests help to confirm stationarity and find any remaining autocorrelation. For the last documented Singapore 2026/LEC session, the chosen stint included 22 uninterrupted laps on SOFT tyres18 used for training and four set aside for testing. With drift, the chosen model was ARIMA(0,1,1), and its stated AIC value was 31.502. The linear-trend baseline showed the lowest RMSE (0.459 seconds) and MAE (0.403 seconds) over the four-lap testing period, therefore exceeding the chosen ARIMA model (RMSE 0.587 seconds; MAE 0.536 seconds). This result is important: picking a model depending on in-sample AIC does not guarantee best out-of-sample prediction performance. The report describes the approach, results, execution specifics, limits, and scope for improvement.

1. Introduction

1.1 Background

Tyre performance is a huge strategic factor in Formula 1. Tyres gradually change during use because of heat and mechanical stresses, and their grip and consistency can deteriorate. Teams monitor lap times, tyre condition, temperatures and other qualities to assess performance. Publicly accessible lap records are less complete than a team’s engineering data, so this project uses lap time as an accessible signal for a compact statistical study.

1.2 Problem statement

Given the sequence of lap times from a reasonably clean tyre stint, can a time-series model forecast the lap times of later laps, and do those predictions provide useful evidence about the direction and magnitude of performance change?

1.3 Aim and objectives

- Retrieve Formula 1 session and lap information programmatically with FastF1.
- Create a cleaner lap-time sequence by removing pit-lane, non-green-flag and inaccurate observations.
- Select a stint and convert timedeltas into numeric seconds.
- Assess differenced-series stationarity and fit candidate ARIMA models.
- Compare ARIMA forecasts with simple, interpretable baselines.
- Evaluate prediction error and inspect residual autocorrelation.
- Interpret what the results can and cannot establish about tyre degradation.

1.4 Scope

The scope is single driver, single session and single stint forecasting. The current code does not estimate tyre wear in millimetres, grip coefficient, carcass temperature or remaining tyre life. It also does not use a multivariate model containing fuel load, weather, traffic or telemetry. It is best understood as a baseline statistical forecasting project and an investigation of lap-time trends.

2. Understanding F1 Tyre Degradation

2.1 What is tyre degradation?

Tyre degradation is the loss of tyre performance during use. Its causes include wear and tear, thermal stress, changes in rubber properties and the way the tyre is loaded through braking, acceleration and cornering. Different compounds and track layouts produce different wear and thermal profiles. A softer compound may provide strong initial grip but can be more sensitive to degradation; actual behaviour depends on the specific tyre construction, conditions and race.

2.2 Lap time as a proxy

Lap time is the total time required to complete a lap. If a tyre becomes less effective, the driver may need to reduce cornering speed or may struggle to maintain traction, increasing lap time. However, the relationship is not one-to-one. A car can become faster as fuel burns off, the track can improve, a driver can manage tyres, and traffic or yellow flags can distort individual laps. A positive lap-time slope is therefore consistent with worsening pace, but it is not proof that tyre wear alone caused the increase.

3. Data and Tools

3.1 Data source: FastF1

FastF1 is a Python package that provides access to Formula 1 timing, lap and telemetry data. In this project, fastf1.get_session(YEAR, EVENT, SESSION_TYPE) identifies the race session, and session.load(telemetry=False, weather=False) loads the session data needed for lap analysis without explicitly loading telemetry or weather. The script enables a local cache directory so previously downloaded session data can be reused and repeated runs can be faster.

3.2 Case-study configuration and provenance

The numerical results in this report are from the 2026 Singapore Grand Prix, driver LEC, stint 3, SOFT compound data.

Item	Singapore / LEC run
Event	2026 Singapore Grand Prix
Driver	LEC
Stint selection	Stint 3
Compound	SOFT
Clean laps	22
Train / test	18 / 4

4. Data Preparation and Lap Filtering

4.1 Driver and lap selection

The driver lap table is selected using session.laps.pick_drivers(DRIVER). The script then applies three filters: pick_wo_box() excludes laps associated with the pit lane; pick_track_status('1') retains laps marked as green-flag running; and pick_accurate() removes laps FastF1 identifies as inaccurate. These filters reduce obvious disruptions but cannot remove every confounder, such as traffic or fuel-load changes.

4.2 Minimum sample size

The script stops if fewer than ten selected laps remain. ARIMA models estimate parameters from temporal patterns, so extremely short sequences provide weak evidence and unstable parameter estimates. Ten observations is only a pragmatic lower bound, not a guarantee of adequate statistical power. The Singapore case study used 22 clean laps, which were split into 18 training observations and four test observations.

5. Experimental Design and Evaluation

5.1 Train/test split

The last episode of the show is presented as test material. With n_test = max(3, len(ts) // 5), the test set is either three observations or one-fifth of the total observations, whichever is greater. Run in Singapore, twenty-two observations produced four held-out laps and eighteen training laps. The models' predictions are compared with the actual test lap times fit without utilizing the test observations.

For projecting, a time-ordered split is fitting as it simulates past lap prediction from future laps. Randomly mixing observations would expose temporal data and would not reflect the actual prediction work.

5.2 Baselines

- Naive forecast: repeats the final training lap time for every test lap. It is a useful minimum benchmark and tests whether a complex model improves on simply assuming no change.
- Linear trend (OLS): fits a straight line to training lap times using the observation index, then extrapolates the line across the test horizon. It is simple and interpretable, but assumes a constant trend.
- ARIMA selected by AIC: uses the candidate model with the lowest training AIC.
- Alternative ARIMA(1,0,0) + trend: an additional model that represents a stationary autoregressive process with a deterministic trend, included if fitting succeeds.

7. Results: Singapore GP, LEC

7.1 Run summary

Measure	Reported result
Event / driver	2026 Singapore Grand Prix / LEC
Stint / compound	Stint 3 / SOFT
Clean laps used	22
Training / test	18 / 4
OLS net degradation slope	+0.077 seconds per lap
ADF p-value after first differencing	Approximately 0.0000
Selected model	ARIMA(0,1,1) + drift

Selected model AIC	31.502
Ljung–Box test	Lag 5; p = 0.8861

The OLS slope of +0.077 s/lap indicates that the fitted linear trend across the training laps rises by roughly 77 milliseconds per lap. This is a net pace trend, not a direct physical measurement of tyre wear. Fuel burn-off and other effects may offset or amplify the trend.

7.2 Held-out forecast performance

Model	RMSE (s)	MAE (s)	Interpretation
Naive	0.531	0.431	Repeats the final training lap time.
Linear trend (OLS)	0.459	0.403	Best error on this reported test window.
ARIMA(0,1,1) + drift	0.587	0.536	Selected by AIC, but worst test error among these four.
ARIMA(1,0,0) + trend	0.466	0.413	Close to the linear baseline; better than naive on this test.

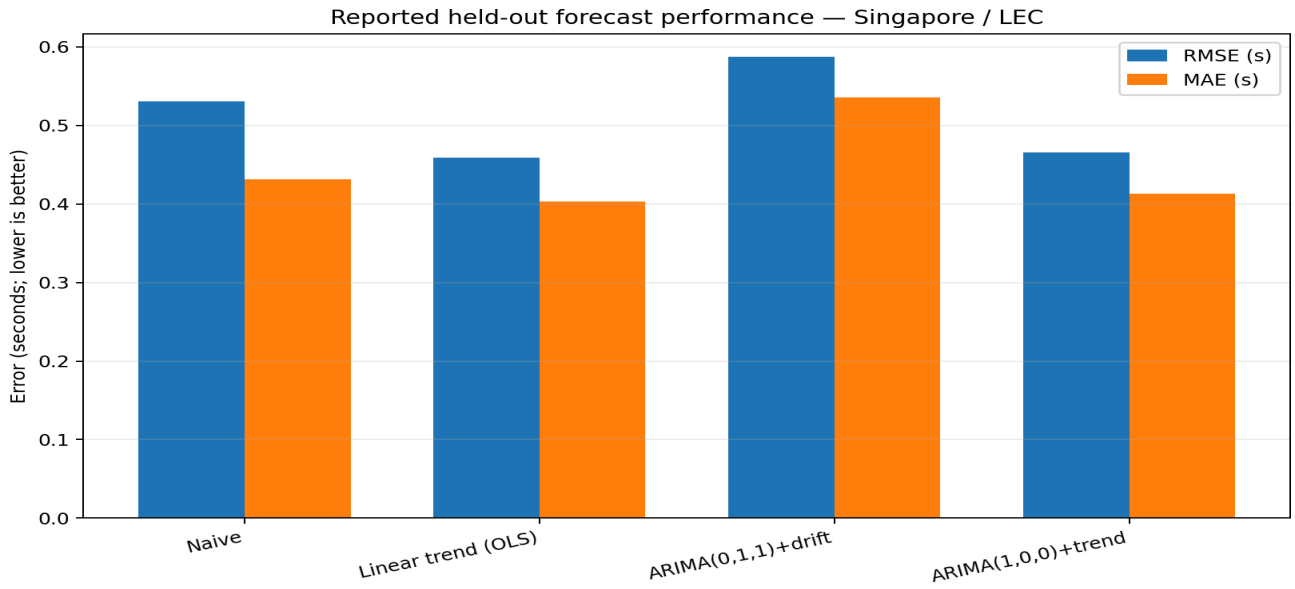


Figure 1. Reported held-out RMSE and MAE for the Singapore / LEC run. Lower bars indicate lower prediction error.

7.3 What the results mean

The linear-trend model beat the AIC-selected ARIMA(0,1,1) + drift model by obtaining the lowest test RMSE (0.459 s) and MAE (0.403 s). This emphasizes that RMSE and MAE assess prediction errors while AIC measures model fit, hence the lowest AIC does not ensure the highest projecting accuracy. The four-lap test set constraints on the dependability of this comparison. Likewise, rolling-origin evaluation indicated that the naive model's RMSE (0.503 s) was lower than that of linear trend (0.522 s) and ARIMA (0.539 s), hence implying that in this specific run ARIMA did not outperform simpler baselines.

7.5 Forecast plot interpretation

The script's prediction number is meant to highlight training lap times, actual kept-out lap times, ARIMA point projections, linear-trend predictions, and the ARIMA 95% confidence range. When analyzing the plot, compare the projections with the held-out real data rather than just with the training fit. A forecast line that seems smooth or likely may not be correct; the numerical holdout data should drive the comparison.

Since this study was put together, the precise lap-by-lap prediction estimate from the Singapore run was not accessible as an image asset; hence no synthetic lap-time points have been generated or shown as real data. Figure 1 shows the model metrics earlier described. Run the script with YEAR = 2026, EVENT = 'Singapore', DRIVER = 'LEC', and STINT_NUMBER = 3 to get the real forecast-versus-observed lap-time graph; then, after checking the displayed metrics, paste the produced forecast_vs_actual.png.

8. Detailed Explanation of the Python Implementation

On GitHub the source code is presented apart. This part clarifies its primary logical units without presenting the whole script.

8.1 Imports and warning suppression

The script imports tools for file management, arithmetic analysis, data processing, plotting, FastF1, ARIMA modelling, and the Ljung–Box residual test as well as for the ADF stationarity test. Though it could obscure convergence or model-specification problems during development, warning suppression helps to keep the output short.

8.2 Configuration and caching

The session and data selection are defined by YEAR, EVENT, SESSION_TYPE, DRIVER, and STINT_NUMBER; SHOW_DIAGNOSTICS regulates auxiliary output. The script uses fastF1 caching to reuse downloaded data and makes a cache folder.

8.3 Loading and cleaning laps

FastF1 loads the session and the script filters the laps of the chosen driver then summarises them by stint. If there aren't any clean laps left, execution stops. Otherwise it picks the given stint or defaults to the one with the most clean laps, deletes the first lap, runs the quick-lap filter, and sorts by lap number.

8.4 Creating the series and split

Lap numbers are kept for charting, lap times are changed from timedeltas to seconds, and the tyre compound is noted. While earlier observations constitute the training set, the last n_test observations shape the test set. Training data just helps to fit models.

8.5 Linear slope and baseline predictions

np.polyfit calculates the slope and intercept of a linear trend fitted to training lap times. The linear baseline projects the fitted trend while the naive baseline repeats the previous training value. These provide criteria for assessing ARIMA.

8.6 ADF and candidate model selection

The first-differenced series is subjected to the ADF test to evaluate stationarity. Training data is used to fit candidate ARIMA models, their orders and AIC values noted. If all fail, execution stops; failed fits are skipped. The applicant chosen has the lowest AIC.

Methodological note: The provided code uses ADF on the whole series, including test data. Though ARIMA factors are fitted just on training data, this test evaluates data from the test period. For leak-free analysis, run ADF on training data just.

8.7 Confidence interval and forecast generation

The chosen ARIMA model is fitted to the training set. get_forecast(steps=n_test) makes predictions and confidence ranges for the test horizon. If fitting fails, an optional ARIMA(1,0,0) trend is also assessed and left out without halting execution.

8.8 Table of scoring and results

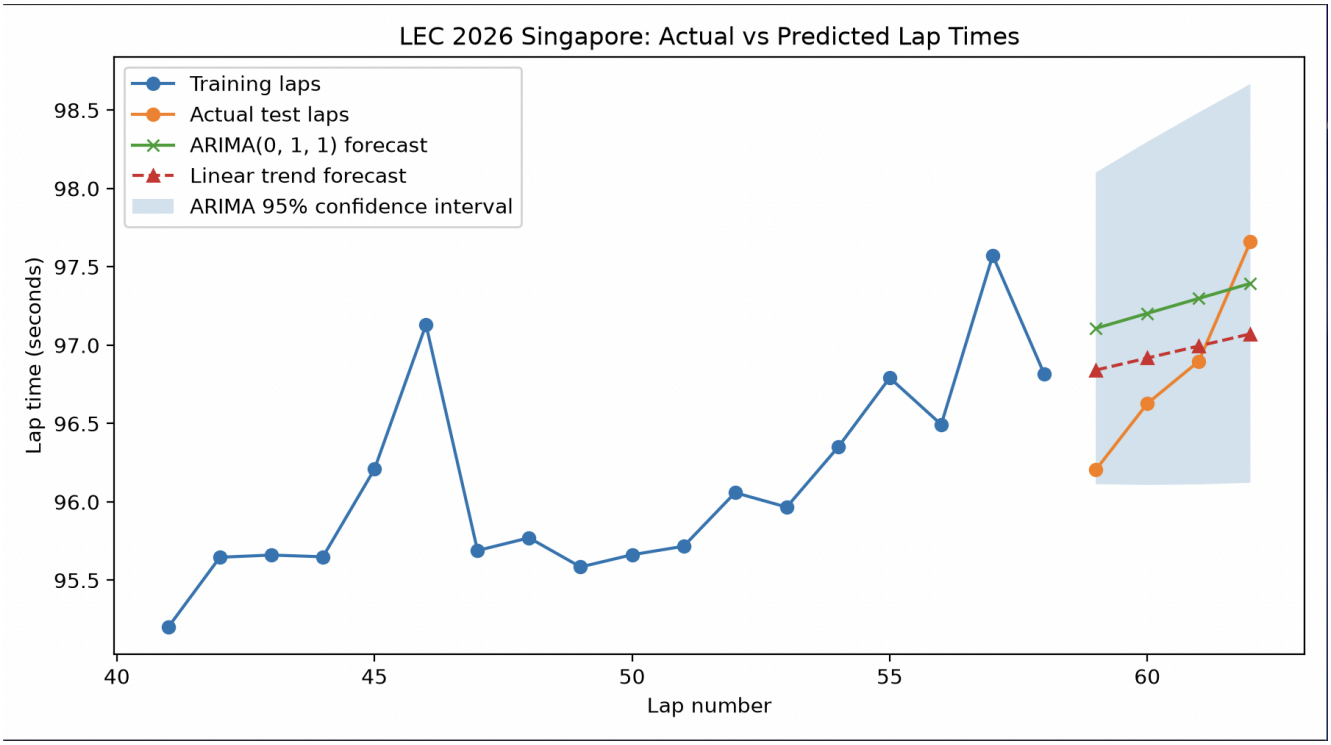
The scoring method evaluates forecasts against real testing values. MAE quantifies mean absolute error and RMSE assesses the square root of mean squared error. A pandas DataFrame holds results rounded to three decimal points. The script determines the top model for every metric; they might differ as they penalise mistakes differently.

8.9 Residual analysis

The script uses (p+q) as the Ljung–Box model-degree correction and trims starting residuals based on the differencing order. Based on residual length, it chooses a test lag and manages failures by reporting the diagnosis as unavailable. A short series might not provide enough residuals for a proper test; therefore, overt lag validation would increase robustness.

8.10 Output and plotting

The console shows lap count, OLS slope, ADF p-value, chosen ARIMA order, AIC, forecast horizon, model scores, and Ljung–Box p-value. Optional diagnostics include candidate AIC values and the full model summary. The plot saved as forecast_vs_actual.png contrasts training and test observations with linear and ARIMA forecasts, including the ARIMA confidence interval.




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F1 TYRE DEGRADATION FORECAST
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Race: 2026 Singapore | Driver: LEC
Stint: 3 | Compound: SOFT | Clean laps: 22
OLS net degradation slope: +0.077 s/lap
ADF p-value after first differencing: 2.059e-06
Selected model: ARIMA(0, 1, 1) + drift
Selected model AIC: 31.502
Forecast horizon: 4 held-out laps

FORECAST PERFORMANCE
                       RMSE (s)  MAE (s)
Model
Naive                     0.531    0.431
Linear trend (OLS)        0.459    0.403
ARIMA(0, 1, 1) + drift    0.587    0.536
ARIMA(1,0,0) + trend      0.466    0.413

DIAGNOSTICS
Ljung-Box p-value (lag 5): 0.8861
Lowest test RMSE: Linear trend (OLS)
Lowest test MAE: Linear trend (OLS)

Forecast plot saved as forecast_vs_actual.png
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9. Conclusion

This work carries out a whole baseline time-series forecasting pipeline for Formula 1 lap times: getting lap records with FastF1, filtering laps, choosing a stint, evaluating differencing, contrasting candidate ARIMA orders by AIC, projecting a held-out period, benchmarking against naive and linear-trend forecasts, determining RMSE and MAE, and examining residual autocorrelation.

Previously seen in the 2026 Singapore Grand Prix run for LEC, stint 3 on SOFT tyres included 22 chosen clean laps. The training-set OLS trend was +0.077 s/lap; the differenced ADF p-value was around 0.0000; and ARIMA(0,1,1) with drift was chosen with AIC 31.502. Nonetheless, at RMSE 0.459 s and MAE 0.403 s—as

opposed to 0.587 s and 0.536 s for the chosen ARIMA model—the linear-trend baseline showed the best reported held-out performance. This shows why straightforward baselines and unseen data help to judge prediction models.

Though its findings should not be overemphasized as a straight-on measure of tyre wear, the study is effective as an educational and replicable forecasting exercise. More stints, more explanatory data, regular validation, and clear modeling of the other elements influencing lap time would help to provide more convincing conclusions.

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